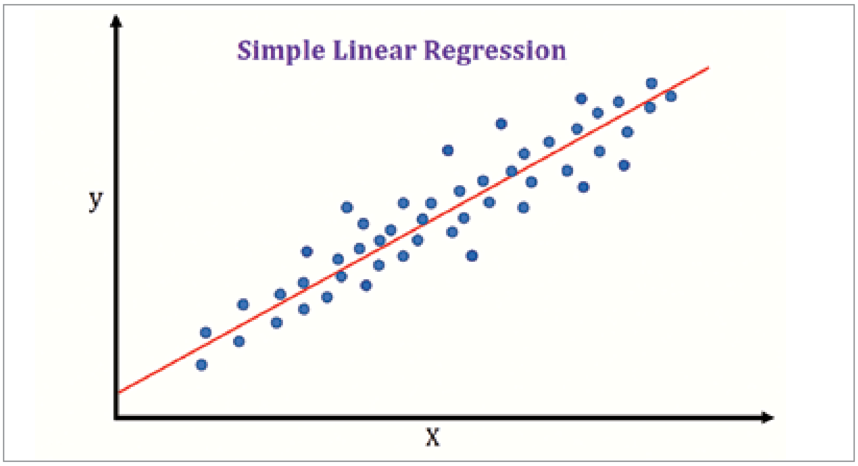

Fig. 5: Simple Linear Regression algorithm (Courtesy: Google Images)

tive linear relationship, then a line can be drawn to predict the outcome of a given input variable depending on its relation with the independent variable. With each iteration the line can be adjusted, until the probability of errors is significantly reduced and the algorithm is optimised as shown in Fig. 5.

The big players

The autonomous vehicle industry has an interesting mix of tech giants and startups racing against time to be first. While all of them are working to build self-driving vehicles, they don't all share the same goals.

1. Waymo. A subsidiary of Google, Waymo is leading the race in developing self-driving cars. Having driven over 32 million kilometres (20 million miles) on public roads, Waymo has spent the past few years testing its technology in 25 different cities across the USA. This is way higher than its competitors such as Baidu whose vehicles so far have driven over 1.6 million kilometres (1 million miles). Waymo intends to use its driverless technology as a ride hailing service, similar to Uber. You get to download an app using which you can hail a Waymo ride. This service is currently in its beta phase, available only at Phoenix Metro Area in Arizona, USA.

2. General Motors (GM). General Motors plans to launch their own taxi service using autonomous vehicles. With years of reputation and outrageous amounts of financial backing, GM requires no formal introduction in the automobile industry. Unlike Waymo, GM engineers its cars completely by themselves. Even the sensors used in the car are either manufactured by them or the subsidiaries they own.

3. Baidu. Baidu is the Chinese equivalent of Google who is heavily invested in building its own autonomous vehicle technology. The major difference between Waymo and Baidu is that, unlike Waymo, which is focused more on autonomous cars, Baidu is focused on building a commercial autonomous bus service that will run across China.

4. Yandex. The Russian equivalent of Google, Yandex has built its own cab hailing service using autonomous vehicle technology. It has been driven for over 8 million kilometres (5 million miles) and is predicted to become a household name in the Russian landscape within the next five years.

Why is it taking so long?

With all the big players mentioned in the previous section pouring in money, manpower and insane amounts of computational power behind the technology, one can't help but wonder why it is taking so long for self-driving cars to become an everyday phenomenon on the streets?

We have the tech, we have the big data, and we have the deep learning models in place. So, what is the problem? Short answer: The technology is not ready, nor are we.

It is a misconception to assume that the technology is ready. Even with millions of kilometres driven with the help of amazing gadgets there are certain pitfalls that we are yet to overcome. Additionally, we don't just require the technology, but also the infrastructure, legislation, and a change of mindset to handle this massive transformation.

Make no mistake, in the automobile industry, the impact of a self-driving car is no less than the invention of the wheel. It completely transforms not just how transportation is used, but many different aspects of our lives.

Technical limitations

Listed below are some of the major reasons that are causing the delay in making this technology available to everyone.

1. Range, speed, and resolution of ultrasonic sensors. While we do have faster and more agile sensors like radar and lidar, a large part of the data that is collected by autonomous vehicles is still dependent on ultrasonic sensors. A major technical snag faced by ultrasonic sensors is that their efficiency massively reduces as the vehicle accelerates beyond a certain point. So, the faster a vehicle, the less reliable the ultrasonic sensor becomes.

Additionally, ultrasonic sensors have a range of only two metres, which provides barely enough time for the collection of data, its computation, and critical decision making. Another drawback of ultrasonic sensors is the resolution which is way low as compared to other sensors such as radar. This creates ambiguity in data and leads to errors in decision making.

2. Range, resolution, and detection issues with radar. Radars are certainly an improvement on ultrasonic sensors but come with their own set of shortcomings. One of the major issues with radars is the false alarms raised by the sensor due to surrounding metal objects. The sensor can often confuse a metal object with another vehicle when, in reality, the object could be anything. To overcome this error, a lot of computational power has to be wasted on correlating the radar data with the data of other sensors to understand false flags.

The range of a radar varies from 5 metres to 200 metres, depending on the quality. But even with the most expensive radar sensors, the resolu-